

OpenCPU SDK

STK API Notes

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Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

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Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Dong Nengxiang

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document introduces STK-related APIs in the OpenCPU SDK.

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2 Data Structure

2.1 typedef enum

```
typedef enum nwy_sat_onoff_type
{
    NWY_SAT_TYPE_OFF = 0,
    NWY_SAT_TYPE_ON = 1,
    NWY_SAT_TYPE_MAX,
}nwy_sat_onoff_type_e;
```

```
typedef enum nwy_sat_pci_type
{
    NWY_SAT_PCITYPE_HANDLE_BY_TE = 0,
    NWY_SAT_PCITYPE_HANDLE_BY_ME = 1,
    NWY_SAT_PCITYPE_NO_OTHER,
}nwy_sat_pci_type_e;
```

2.2 typedef struct

```
typedef struct nwy_sat_notify_info_type
{
    nwy_sat_pci_type_e pcitype;
    uint16_t nDataLen;
    char pData[540];
}nwy_sat_notify_info_type_t;
```

3 APIs

The interface function definition can be found in `nwy_sat.h`, and is used to implement STK-related functions.

3.1 `nwy_sat_set_on_off`

Format	<code>nwy_error_e nwy_sat_set_on_off(nwy_sat_onoff_type_e switch_type);</code>
Description	Enables or disables the STK function.
Parameter	<code>nwy_sat_onoff_type_e</code> : Enable or disable the function.
Return Value	Success: <code>NWY_SUCCESS</code> Failure: another enumeration value

3.2 `nwy_sat_get_on_off`

Format	<code>int nwy_sat_get_on_off(void);</code>
Description	Retrieves the STK enable status.
Parameter	Void
Return Value	Success: <code>NWY_SUCCESS</code> Failure: another enumeration value

3.3 `nwy_sat_terminal_response`

Format	<code>nwy_error_e nwy_sat_terminal_response(char *data);</code>
Description	Forwards the mobile phone's command execution result to the SIM card via STKTR.
Parameter	<code>data</code> : Hexadecimal string of the STK response.
Return Value	Success: <code>NWY_SUCCESS</code> Failure: another enumeration value

3.4 nwy_sat_envelope_command

Format	nwy_error_e nwy_sat_envelope_command(char *data)
Description	Sends signaling data from the mobile device to the SIM card.
Parameter	data: Hexadecimal string of the signaling data.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.5 nwy_sat_set_profile

Format	nwy_error_e nwy_sat_set_profile(char *data);
Description	Sets the profile.
Parameter	data: Hexadecimal string for profile configuration.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.6 nwy_sat_get_profile

Format	nwy_error_e nwy_sat_get_profile(char *data);
Description	Retrieves the profile.
Parameter	data: Hexadecimal string for profile configuration.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.7 nwy_sat_rcv_data

Format	void nwy_sat_rcv_data(nwy_sat_notify_info_type_t *sat_data)
Description	Retrieves data reported by STKPCI.
Parameter	URC format: +STKPCI:<pcitype>,<string> <pcitype>: 0 The SAT command is handled by TE. 1 The SAT command is handled by ME. 2 No other command (end of session) <string>: Binary raw data structure of the unsolicited command.

Return Value	Success: NWY_SUCCESS Failure: another enumeration value
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